A six-layer bound for quartic graphs of girth five and the minimum order at diameter twenty-two

Computational reconstruction of a girth–diameter bound

Abstract

We prove that six consecutive nonsingleton distance layers of a simple quartic graph of girth at least five contain at least twenty-two vertices. The last local cases are excluded by twenty-seven explicit Boolean refutations, whose formulas are reconstructed and whose proofs are checked without a satisfiability solver. Combining this local bound with boundary balls and a mixed-radius packing inequality shows that a quartic graph of girth at least five with a vertex of eccentricity twenty-two has at least ninety-five vertices. An included ninety-five-vertex graph has degree four, girth five and diameter twenty-two. Thus $n(4; 5, 22) = 95$. No claim about the number of nonisomorphic minimum graphs is needed.

1 Statement, notation and antecedents

For fixed degree k, girth g and diameter d, write $n(k; g, d)$ for the minimum order of a connected simple k-regular graph with those parameters. Table 5 of Cambie, Goedgebeur, Jooken and Van den Eede [1] records $n(4; 5, 22) = 95$ and at least one such graph. We give a lower bound for a slightly larger domain and separately verify a graph attaining it.

Theorem 1. *Let G be a connected simple 4-regular graph of girth at least five. If some vertex has eccentricity 22, then $|V(G)| \geq 95$.*

Fix a root o and write $V_i = \{x : d_G(o, x) = i\}$. The symbols V_i always refer to these original distance layers. A ball taken in an induced subgraph H is written $\mathrm{B}_r^H(x)$, and its sphere is $\mathrm{S}_r^H(x)$. Every edge changes the original index by at most one. Consequently a ball of radius r centred at original index j is supported on indices $j - r, \ldots, j + r$, even when the ball is taken in an induced subgraph. We never identify reversed original layers with distance layers from a new root.

The Moore-tree argument gives $|\mathrm{B}_2(x)| = 1 + 4 + 12 = 17$ whenever all relevant vertices have degree four: triangles and quadrangles are exactly the obstructions to distinctness. Five consecutive nonempty original layers also have mass at least 17: take a geodesic through their middle layer and its radius-two ball. The distance-layer framework and these Moore interval bounds are reused from [1]. The elementary methods carry no claim of priority. Detailed hand proofs used below are included as companion notes, rather than replaced by a numerical search.

2 The local six-layer theorem

Theorem 2. *Six consecutive original distance layers, each containing at least two vertices, have total mass at least 22.*

The hand arguments in `premises/six_layer_mass_bound.rst` and `premises/six_layer_twenty_equality_exclusion.rst` establish mass at least 21. We describe the remaining equality cases and the exact interface to the finite certificates.

Suppose the six layers have total mass 21, and number them locally $0, \dots, 5$. They begin after the root layer, which is a singleton. Choose a geodesic across them, with adjacent centres u in layer 2 and v in layer 3. Their other neighbours are A_1, A_2, A_3 and B_1, B_2, B_3, labelled so that A_1 is in layer 1 and B_1 in layer 4 on the geodesic. No A–B edge is possible. A common second neighbour C_{ij} has parents A_i, B_j; at most one exists for each cell ij, and it cannot have a second parent on either side. These assertions follow respectively from the absence of triangles or quadrangles. If m grid cells are occupied, then

$$|\mathrm{B}_2(u) \cap \mathrm{B}_2(v)| = 8 + m, \qquad |\mathrm{B}_2(u) \cup \mathrm{B}_2(v)| = 26 - m. \tag{1}$$

For each missing cell ij, name a private neighbour U_{ij} of A_i and a private neighbour V_{ij} of B_j. All these private vertices are distinct. Each of the eight centre or parent vertices is already saturated by its four prescribed neighbours.

Cell 11 is missing: a common neighbour would connect original indices whose difference is three by a path of length two. Thus $m \leq 8$. Equation (1) immediately settles $m \leq 4$. At total mass 21 the only remaining possibilities are

m	8	7	6	5
vertices outside the displayed union	3	2	1	0.

The $m = 8$ case is excluded in `premises/six_layer_equality_eight_cells.rst`. For $m = 7$ the two missing cells either share a row or column, or lie in distinct rows and columns. Their respective exclusions are in `premises/six_layer_adjacent_holes_proof.rst` and `premises/six_layer_equality_matching_holes.rst`. The former is the corrected full proof, including its permitted five-cycle subcase. These hand arguments do not assume that reversing the interval preserves the original predecessor axiom.

2.1 A necessary Boolean model for $m = 5, 6$

For $m = 5$ there are exactly twenty-one displayed vertices. For $m = 6$ add one vertex X. Private names can be chosen so that the six geodesic vertices are

$$U_{11}, A_1, u, v, B_1, V_{11},$$

in local layers $0, 1, 2, 3, 4, 5$, respectively. For each vertex introduce six Boolean layer variables with exactly one true entry. The prescribed edges are constants true; all other edges incident to the saturated eight vertices are false. All remaining unordered pairs receive an edge variable. The formula requires:

1. the six anchors above and at least two vertices in every layer;
2. endpoints of an edge to have indices differing by at most one;
3. degree at most four everywhere, and degree four at local indices $1, \dots, 4$;
4. degree at most three inside the interval at local index zero;
5. for each vertex at index $h > 0$, a neighbour at index $h - 1$;
6. at most one common neighbour for any pair, and none for an adjacent pair.

There is no successor condition. At index zero there is no positive internal-degree requirement: its compulsory predecessor is outside the interval. Five-cycles remain allowed. The last condition excludes quadrangles and triangles, not pentagons.

Every actual graph fragment satisfying our hypotheses gives a satisfying assignment. Assign its original indices and its induced adjacency, then evaluate the equivalence-defined

Boolean gates. Interior degrees are complete because an edge cannot leave the interval from an interior index. At index zero the original nonroot layer supplies a predecessor outside. All other requirements were just established. The deterministic formula source and its Boolean gate library are included unchanged; only the latter's elementary gates are used, not its unrelated graph builder. A full encoding proof is `premises/proof.rst`.

2.2 Complete coverage and checked refutations

The available label symmetries are independent swaps of rows $2, 3$ and columns $2, 3$. They fix cell 11, all geodesic anchors and all layer indices. They transport private names with their parents. Hence this $C_2 \times C_2$ action preserves every condition above. Neither transpose nor reversal is used. There are

$$\binom{8}{2} = 28 \quad (m = 6), \qquad \binom{8}{3} = 56 \quad (m = 5)$$

hole sets containing 11, partitioned into 10 and 17 orbits. `data/girth_diameter_bounds/hole_pattern_partition.json` lists every member and every one of its four images; the companion coverage proof justifies the action.

For each of the 27 representatives, the package contains a deterministic DIMACS formula, its primary-variable map and a complete proof stream. The replay reconstructs the entire formula and map byte for byte, then checks the saved proof with the included reverse-unit-propagation (RUP) checker. Each added clause must be justified; an explicit empty-clause derivation is required. The checker retains the fixed original-unit consequences and a historical fallback, whose soundness arguments are in `premises/root_units.tex` and `premises/root_units_active.tex`. Its sources and imported source are included; a local C++17 compiler rebuilds the executable. The solver's original answer is not a premise, and no solver is required for replay. The exact 27 proofs contain 43 029 checked additions in total.

Thus each necessary formula is contradictory. The label action covers every $m = 5, 6$ hole pattern, so all equality-21 cases are excluded. Together with the hand lower bound 21, this proves Theorem 2.

3 Boundary facts and singleton layers

If an internal original layer is a singleton $\{w\}$, its two cuts are positive even integers. Indeed every degree is even, both cuts are nonempty, and their sum is four. They are therefore both two. The induced prefix and suffix including w are connected graphs in which w has degree two and every other vertex has degree four; call either one a *cap*. The induced section between consecutive singleton layers has two degree-two boundary vertices and all other degrees four. Connectivity follows from original root geodesics.

We use the following hand-proved facts from the included notes.

Lemma 3. *Under the preceding degree and girth hypotheses:*

1. *Every cap has at least* 20 *vertices.*
2. *A boundary ball* $\mathrm{B}_3(w)$ *has at least* 15 *vertices when no other exceptional boundary occurs nearby. A boundary ball* $\mathrm{B}_4(w)$ *has at least* 19 *vertices if all its other vertices have degree four, the induced graph has even degrees, and a vertex at distance at least five exists.*
3. *If* $|G| \leq 94$ *and* o *has eccentricity* 22*, the complete set of internal singleton indices is one of* $\varnothing, \{4\}, \{18\}, \{4, 18\}$.

The cap proof is `premises/singleton_cap_twenty_bound.rst`; its earlier degree-two cap argument is retained in `premises/singleton_cap_bound.tex`. The boundary proof is `premises/boundary_radius_four.rst`. The position reduction uses `premises/singleton_layer_section.tex`, `premises/singleton_pattern_bound.rst`, `premises/singleton_free_section_bounds.rst`, and the cap and six-layer bounds above. The stages reduce the necessary nonempty patterns to 66, then 14, then 12, then 5, and finally the three displayed nonempty sets. The last step already follows from mass 21, in `premises/six_layer_twenty_equality_exclusion.rst`. These are reductions of the complete set of singleton positions, not selections of convenient individual singletons.

4 A mixed-radius boundary inequality

Lemma 4. *Under the radius-four hypotheses of Lemma 3,*

$$|\mathrm{B}_3(w)| + |\mathrm{B}_4(w)| \geq 35. \tag{2}$$

Proof. The first two spheres have sizes two and six. Split the six vertices into triples P, Q according to their first-sphere parents. Their internal edges form a matching of size $m \leq 3$. Every third-sphere vertex has at most one parent in each triple, so $18 - 2m \geq 12$ incidences enter that sphere and $c = |\mathrm{S}_3(w)| \geq 6$.

If $c \geq 7$, then $|\mathrm{B}_3(w)| = 9 + c \geq 16$, and the separate bound $|\mathrm{B}_4(w)| \geq 19$ proves (2). If $c = 6$, the incidence count forces $m = 3$ and two parents for every third-sphere vertex. Label the three matching edges $p_i q_i$, $i = 0, 1, 2$. The six third-sphere labels are exactly the off-diagonal pairs ij; they are independent by the triangle and quadrangle restrictions with the parents. Thus twelve incidences enter the fourth sphere.

A fourth-sphere vertex cannot meet two labels sharing a coordinate, as that would give a quadrangle. It has at most three neighbours in the third sphere. The only possible three-label sets are $\{01, 12, 20\}$ and $\{10, 02, 21\}$. Neither set can repeat at two fourth-sphere vertices, again by a quadrangle. Four vertices in the fourth sphere could therefore receive at most $3 + 3 + 2 + 2 = 10$ incidences. Its size is at least five. Hence $|\mathrm{B}_3(w)| = 15$ and $|\mathrm{B}_4(w)| \geq 20$, proving (2). □

This is a joint inequality. It does not assert a universal boundary radius-four lower bound of twenty.

5 The global packing

Assume for contradiction that $|G| \leq 94$. Lemma 3 gives four alternatives. A ball of radius two at o has size 17 and support $0, 1, 2$. A radius-two ball at any vertex of V_{22} also has size 17 and support $20, 21, 22$, even if V_{22} is a singleton.

If there is no internal singleton, use those two end balls and the disjoint intervals $3, \ldots, 8$, $9, \ldots, 14$, and $15, \ldots, 19$. Their masses are at least $22, 22, 17$. Thus

$$|G| \geq 17 + 22 + 22 + 17 + 17 = 95.$$

If the only internal singleton is at 4, the left cap has size at least 20. In the right cap use its boundary radius-three ball, the whole intervals $8, \ldots, 13$ and $14, \ldots, 19$, and the terminal radius-two ball. Their supports are disjoint, so the right cap has size at least $15+22+22+17 = 76$. The caps share just their boundary, giving $|G| \geq 20 + 76 - 1 = 95$. For the sole singleton 18, the left cap instead contains supports $0, \ldots, 2$, $3, \ldots, 8$, $9, \ldots, 14$ and $15, \ldots, 18$, with sizes at least $17, 22, 22, 15$. The same count applies without reversing a predecessor axiom.

It remains to treat $\{4, 18\}$. Let H be the section on original indices $4, \dots, 18$, with boundaries w, z. Their distance in H is exactly fourteen: an original root geodesic gives such a path, and indices cannot change faster than one per edge. Thus the radius-four hypotheses hold at both ends. Put $L_r = |\mathrm{B}_r^H(w)|$ and $R_r = |\mathrm{B}_r^H(z)|$. Two separate disjoint packings give

$$\begin{aligned} |H| &\geq L_4 + 22 + R_3, && \text{supports } [4, 8], [9, 14], [15, 18], \\ |H| &\geq L_3 + 22 + R_4, && \text{supports } [4, 7], [8, 13], [14, 18]. \end{aligned}$$

Each middle interval is a complete original nonsingleton six-layer interval. The two packings may overlap each other; only the sets within each one must be disjoint. Adding the inequalities and using Lemma 4 gives

$$2|H| \geq (L_3 + L_4) + (R_3 + R_4) + 44 \geq 114, \qquad |H| \geq 57.$$

The caps outside H each have size at least twenty and meet it only at their respective boundaries. Consequently

$$|G| \geq 20 + 57 + 20 - 2 = 95.$$

This exhausts the alternatives and proves Theorem 1.

6 Equality and reproducibility

The included graph6 file `data/girth_diameter_bounds/n95_k4_g5_d22.g6` contains one graph. The property predicate reused from the prior independent positive check computes connectedness and the exact tuple

$$(|V|, \delta, \Delta, \text{girth}, \text{diameter}, \text{bipartite indicator}) = (95, 4, 4, 5, 22, 0).$$

It uses NetworkX 3.6.1. The portable entrypoint compares the predicate's syntax tree to the included accepted reference before applying it. This positive witness and Theorem 1 prove the following.

Corollary 5. $n(4; 5, 22) = 95$*, and there is at least one such minimum graph.*

The portable replay authenticates all files, verifies the 84-pattern cover, reconstructs all 27 formulas and primary-variable maps, rebuilds the unchanged RUP checker, checks every saved refutation, and computes the positive graph's properties. The handwritten premises remain mathematical proofs to be read; the replay is not a proof-assistant formalization. No old graph-search cache, graph generator, SAT solver, or private filesystem path is needed. Instructions and exact software premises are in the companion README. The result concerns this source row only; it neither completes the other source rows nor enumerates the minimum graphs.

References

[1] S. Cambie, J. Goedgebeur, J. Jooken and T. Van den Eede, *On the order-diameter ratio of girth-diameter cages*, arXiv:2511.21144v3, 19 August 2026; DMTCS 28:4, SOFSEM2026 #1, Table 5, physical PDF page 22.